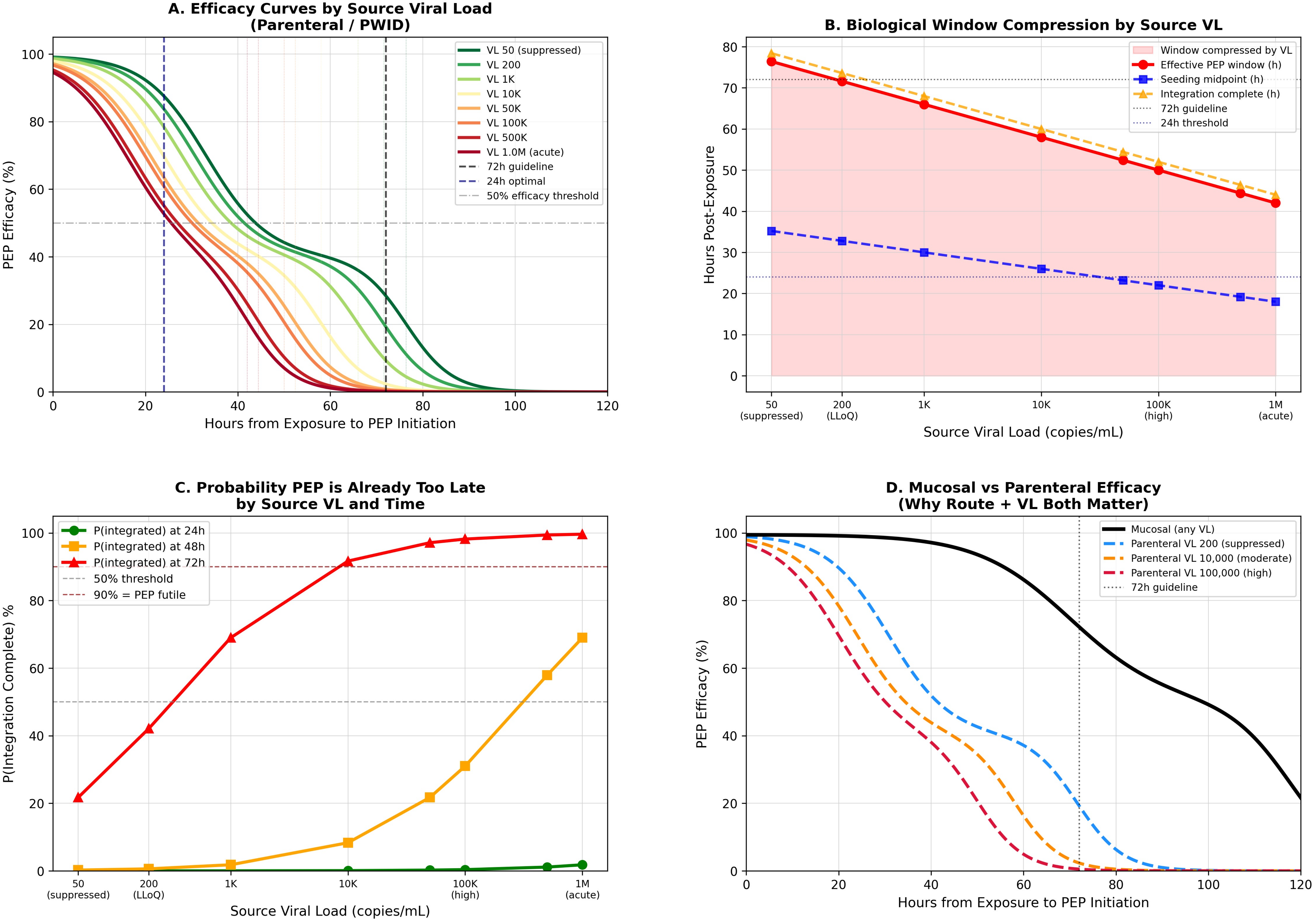
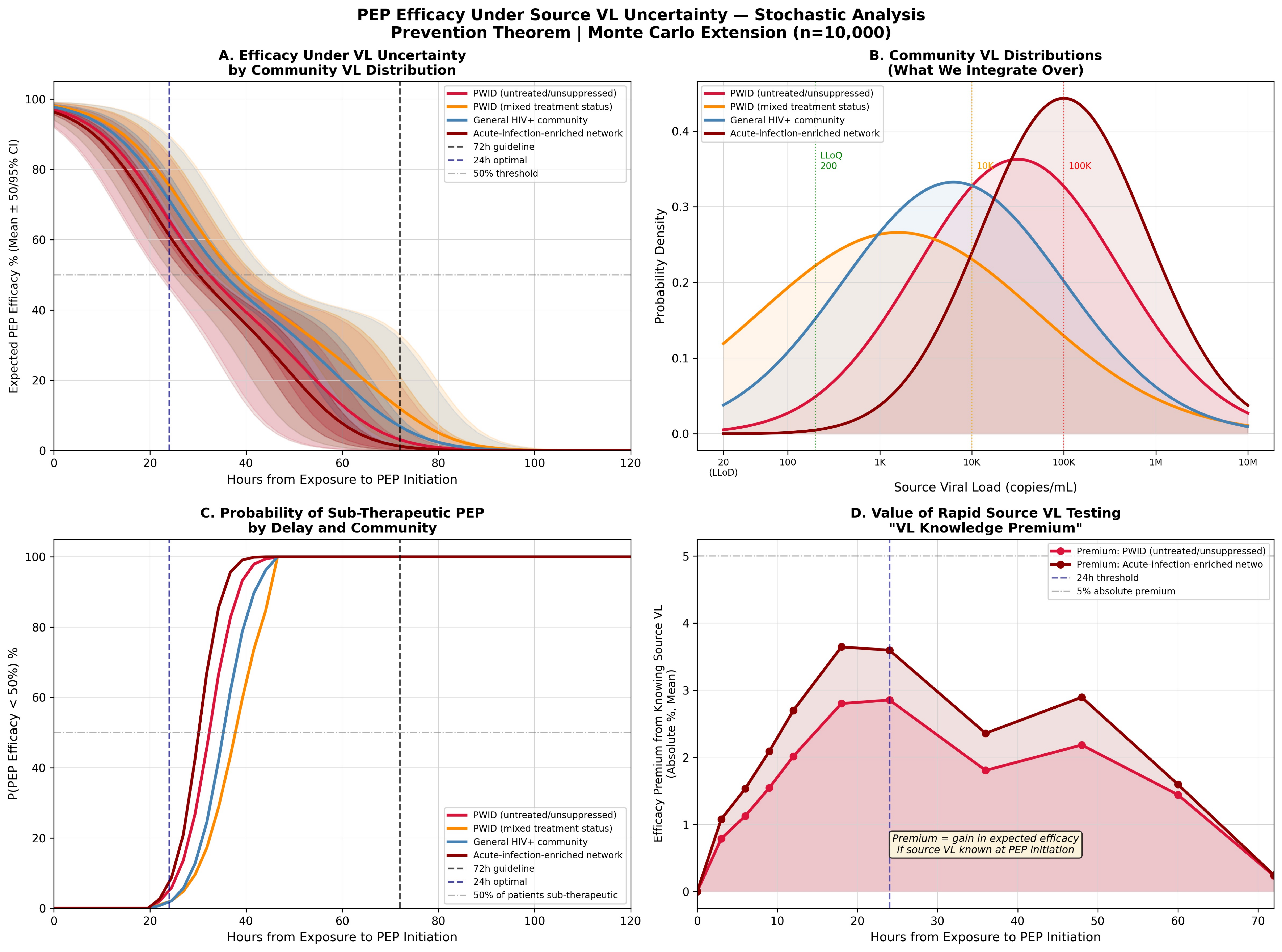
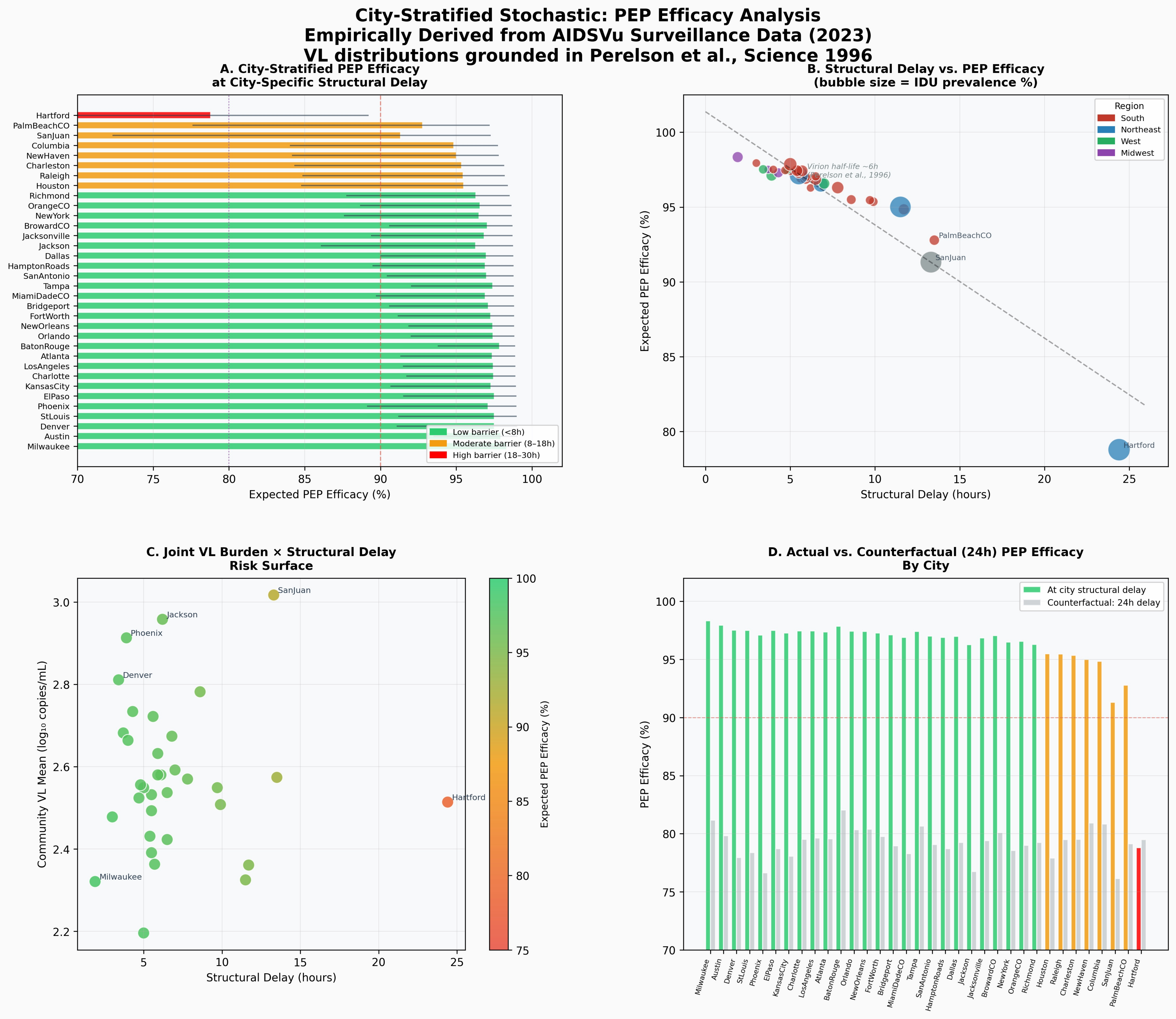
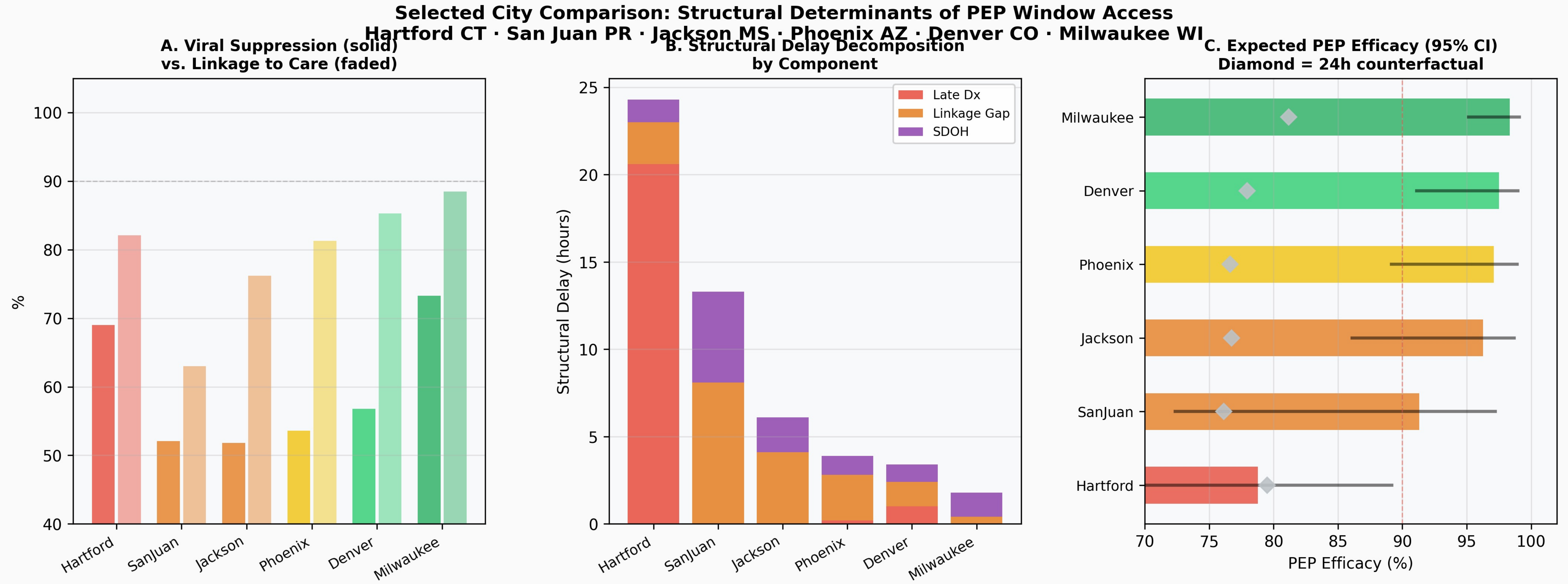
Viral Load as Window Compressor
PEP Parenteral Extension




PEP Efficacy Under Source VL Uncertainty — Stochastic Analysis
Prevention Theorem | Monte Carlo Extension (n=10,000)
A. Efficacy Under VL Uncertainty
by Community VL Distribution
Expected PEP Efficacy % (Mean ± 50/95% CI)
Hours from Exposure to PEP Initiation
PWID (untreated/unsuppressed)
PWID (mixed treatment status)
General HIV+ community
Acute-infection-enriched network
72h guideline
24h optimal
50% threshold
B. Community VL Distributions
(What We Integrate Over)
Probability Density
Source Viral Load (copies/mL)
LLoQ
200
10K
100K
20
(LLoD)
100
1K
10K
100K
1M
10M
C. Probability of Sub-Therapeutic PEP
by Delay and Community
P(PEP Efficacy < 50%) %
Hours from Exposure to PEP Initiation
50% of patients sub-therapeutic
D. Value of Rapid Source VL Testing
"VL Knowledge Premium"
Efficacy Premium from Knowing Source VL
(Absolute %, Mean)
Hours from Exposure to PEP Initiation
Premium: PWID (untreated/unsuppressed)
Premium: Acute-infection-enriched netwo
24h threshold
5% absolute premium
Premium = gain in expected efficacy
if source VL known at PEP initiation



City-Stratified Stochastic: PEP Efficacy Analysis
Empirically Derived from AIDSVu Surveillance Data (2023)
VL distributions grounded in Perelson et al., Science 1996
A. City-Stratified PEP Efficacy
at City-Specific Structural Delay
Hartford
PalmBeachCO
SanJuan
Columbia
NewHaven
Charleston
Raleigh
Houston
Richmond
OrangeCO
NewYork
BrowardCO
Jacksonville
Jackson
Dallas
HamptonRoads
SanAntonio
Tampa
MiamiDadeCO
Bridgeport
FortWorth
NewOrleans
Orlando
BatonRouge
Atlanta
LosAngeles
Charlotte
KansasCity
ElPaso
Phoenix
StLouis
Denver
Austin
Milwaukee
Low barrier (<8h)
Moderate barrier (8–18h)
High barrier (18–30h)
Expected PEP Efficacy (%)
B. Structural Delay vs. PEP Efficacy
(bubble size = IDU prevalence %)
Region
South
Northeast
West
Midwest
Virion half-life ~6h
(Perelson et al., 1996)
PalmBeachCO
SanJuan
Hartford
Expected PEP Efficacy (%)
Structural Delay (hours)
C. Joint VL Burden × Structural Delay
Risk Surface
SanJuan
Jackson
Phoenix
Denver
Hartford
Milwaukee
Community VL Mean (log₁₀ copies/mL)
Structural Delay (hours)
Expected PEP Efficacy (%)
D. Actual vs. Counterfactual (24h) PEP Efficacy
By City
At city structural delay
Counterfactual: 24h delay
PEP Efficacy (%)


Selected City Comparison: Structural Determinants of PEP Window Access
Hartford CT · San Juan PR · Jackson MS · Phoenix AZ · Denver CO · Milwaukee WI
A. Viral Suppression (solid)
vs. Linkage to Care (faded)
%
Hartford
SanJuan
Jackson
Phoenix
Denver
Milwaukee
B. Structural Delay Decomposition
by Component
Structural Delay (hours)
Late Dx
Linkage Gap
SDOH
C. Expected PEP Efficacy (95% CI)
Diamond = 24h counterfactual
PEP Efficacy (%)

Supplementary Text

Finite Prevention Windows for HIV Post-Exposure Prophylaxis: Irreversible Proviral Integration Defines Route-Specific Intervention Limits

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Science submission companion — not for independent distribution

This Supplementary Text contains S1-S6 (mathematical development with complete proofs), S7-S9 (extended methods and parameter tables), and S10 (complete figure inventory with panel-level descriptions and source script mapping).

S1. PROBABILITY SPACE, STATE DEFINITIONS, AND ASSUMPTIONS

Consider a single HIV exposure event e occurring at time t = 0 on probability space (O, F, P) with filtration {F_t}_{t>=0}.

S1.1 Within-host states

Definition S1.1 (Within-host infection states). Z(t) in {0,1,2}:

Z(t)=0 (Susceptible): No productive infection; I(t) < I*.

Z(t)=1 (Seeded): Replicative focus established (I(t) >= I*) but reservoir below irreversibility threshold (R(t) < R*).

Z(t)=2 (Integrated): R(t) >= R*. Absorbing state under Assumption A1.

Definition S1.2 (Transition stopping times). T_{seed} = inf{t>=0 : Z(t)>=1}; T_{int} = inf{t>=0 : Z(t)=2}. We assume T_{seed} <= T_{int} a.s.

Definition S1.3 (Cumulative transition probabilities). $P_{seed}(t)$ = P(T_{seed} <= t); $P_{int}(t)$ = P(T_{int} <= t). Since T_{seed} <= T_{int} a.s.: $P_{seed}(t)$ >= $P_{int}(t)$ for all t >= 0.

S1.2 Assumptions

Assumption A1 (Irreversibility of integration). Once Z(t)=2 the state is absorbing: P(Z(s)=0 | Z(t)=2) = 0 for all s > t.

Biological grounding: Integrated provirus persists for the lifetime of the host cell and propagates to daughter cells via clonal expansion. ART suppresses replication (I->0) but dR/dt = aI -> 0: reservoir is static under ART but does not shrink. No approved therapy achieves eradication.

Assumption A2 (Stage-dependent drug efficacy). e(Z(t)) = e_{max} if Z(t)=0; e_{mid} if Z(t)=1; e_{min}=0 if Z(t)=2. With 1 >= e_{max} >= e_{mid} >= 0.

Assumption A3 (Monotonicity of integration). $P_{int}(t)$ non-decreasing, $P_{int}(0)$=0, lim $P_{int}(t)$=1.

S2. MASTER EQUATION

Proposition S2.1 (Master equation). Under A2:

$$E\_PEP(t) = (1-P\_seed(t))\cdot e\_max + (P\_seed(t)-P\_int(t))\cdot e\_mid + P\_int(t)\cdot e\_min \qquad \text{(S2.1)}$$

Proof.

Partition sample space at time t by Z(t): P(Z=0)=1-P_{seed}, P(Z=1)=P_{seed}-P_{int}, P(Z=2)=P_{int}. Mutually exclusive and exhaustive. By the law of total expectation: $E_{PEP}(t)$ = E[e(Z(t))] = (1-P_{seed})·e_{max} + (P_{seed}-P_{int})·e_{mid} + P_{int}·e_{min}. □

S3. THE FINITE PREVENTION WINDOW THEOREM

Theorem S3.1 (Finite Prevention Window). Under A1-A3, $E_{PEP}(t)$ satisfies: (i) $E_{PEP}(t) <= (1-P_{int}(t))\cdot e_{max}$; (ii) $E_{PEP}(t)$ non-increasing; (iii) $E_{PEP}(t)$->0; (iv) $t_{crit}(h)$ = inf{t: $E_{PEP}(t)$<h} is finite for any h in (0,1).

Proof.

Part (i): Upper bound.

Replace e_{mid} by e_{max} in (S2.1):

$E_{PEP}(t) <= (1-P_{seed})\cdot e_{max}+(P_{seed}-P_{int})\cdot e_{max}+P_{int}\cdot e_{min} = (1-P_{int})\cdot e_{max}+P_{int}\cdot e_{min}$. With e_{min}=0: $E_{PEP}(t) <= 1-P_{int}(t)$.

Part (ii): Monotone decay.

For t1<t2: DS = $P_{seed}(t2)-P_{seed}(t1) >= 0$, Di = $P_{int}(t2)-P_{int}(t1) >= 0$.

$E_{PEP}(t1)-E_{PEP}(t2) = DS\cdot(e_{max}-e_{mid})+Di\cdot(e_{mid}-e_{min}) >= 0$.

Part (iii): Convergence.

By A3, $P_{int}(t)$->1. Since $T_{seed} <= T_{int}$ a.s., $P_{seed}(t)$->1. Limits: $E_{PEP}(t) \to 0\cdot e_{max}+0\cdot e_{mid}+1\cdot e_{min}=0$.

Part (iv): Finite critical time.

By (ii) E_{PEP} non-increasing; by (iii) ->0<h. {t: $E_{PEP}(t)$<h} non-empty, so $t_{crit}(h)$ is finite. □

S4. HAZARD-RATE DECOMPOSITION OF EFFICACY LOSS

Definition S4.1 (Hazard rates). $l_{seed}(t) = f_{seed}(t)/(1-P_{seed}(t))$; $l_{int}(t) = f_{int}(t)/(P_{seed}(t)-P_{int}(t))$.

Proposition S4.1 (Decomposition). Under A1-A3 with absolute continuity:

$$dE\_PEP/dt = -f\_seed(t)\cdot De1 - f\_int(t)\cdot De2 <= 0 \qquad (S4.1)$$

where De1 = $e_{max}-e_{mid}$ and De2 = $e_{mid}-e_{min}$.

Proof.

Differentiate (S2.1): $dE/dt = -f_{seed}\cdot e_{max}+(f_{seed}-f_{int})\cdot e_{mid}+f_{int}\cdot e_{min} = -f_{seed}\cdot De1-f_{int}\cdot De2 <= 0$. □

Remark S4.1. (S4.1) decomposes efficacy loss into a seeding channel (f_{seed}, gap De1) and an integration channel (f_{int}, gap De2). At early times, seeding dominates; as mass accumulates in state 1, the integration channel accelerates terminal efficacy collapse.

S5. ROUTE COMPRESSION AND INOCULUM-MONOTONE HITTING TIMES

S5.1 ODE system

The standard target-cell-limited within-host model:

$$dT/dt = l-dT-bTV,\ \ dI/dt = bTV-dI,\ \ dV/dt = pI-cV,\ \ dR/dt = aI \qquad (S5.1)$$

T = target cells, I = infected cells, V = free virions, R = cells with stable proviral integration.

Lemma S5.1 (Absorbing state). Set A = {R>0} is positively invariant. If R(t0)>0 then R(t)>0 for all t>t0.

Proof.

dR/dt = aI >= 0, so R non-decreasing. Under ART, I->0, dR/dt->0: reservoir is static but does not shrink. □

S5.2 Route-dependent initial conditions

$V^{(m)}(0)$ = 1 virion/mL (mucosal, post-epithelial attenuation) vs. $V^{(p)}(0)$ = 10^3 virions/mL (parenteral, direct intravascular delivery). All ODE parameters are identical across routes.

Lemma S5.2 (Inoculum-monotone hitting times). Solutions of (S5.1) with V(0)=v0 < V_tilde(0)=v_tilde satisfy I_tilde(t)>=I(t) and R_tilde(t)>=R(t) for all t>=0. Therefore T_{int}(v_tilde) <= T_{int}(v0) a.s.

Proof.

The subsystem (I,V,R) is cooperative for fixed T: d(bTV)/dV >= 0, d(pI)/dI >= 0, d(aI)/dR = 0. By comparison theorems for cooperative ODE systems (Smith 1995), solutions are order-preserving in initial conditions. Since V_tilde(0) > V(0): V_tilde(t) >= V(t) and I_tilde(t) >= I(t) for all t >= 0. Then:

R_tilde(t) = a·integral_0^t I_tilde(s) ds >= a·integral_0^t I(s) ds = R(t). So T_{int}(v_tilde) <= T_{int}(v0). □

S5.3 Route compression corollary

Assumption A4 (Stochastic dominance). $P_{int^{(p)}}$(t) >= $P_{int^{(m)}}$(t) for all t >= 0. Follows from Lemma S5.2 under route-specific inocula.

Corollary S5.1 (Route compression). Under A1-A4: (i) $E_{PEP^{(p)}}$(t) <= $E_{PEP^{(m)}}$(t); (ii) $t_{crit^{(p)}}$(h) <= $t_{crit^{(m)}}$(h).

Proof.

(i) Weight on e_{max} = 1-P_{seed}(t) is smaller for parenteral; weight on e_{min}=0 is larger. So $E_{PEP^{(p)}}$ <= $E_{PEP^{(m)}}$.

(ii) Both non-increasing; $t_{crit^{(p)}}$(h) = inf{t: $E_{PEP^{(p)}}$<h} <= inf{t: $E_{PEP^{(m)}}$<h} = $t_{crit^{(m)}}$(h). □

S6. POPULATION-LEVEL PEP EFFICACY BOUND

Definition S6.1. E_bar$_{PEP}$ = E[E_{PEP}(T_{access})] = integral E_{PEP}(t) dF_{access}(t).

Proposition S6.1 (Population-level bound). Under A1-A4:

E_bar_PEP <= F_access(t_crit)·e_max + (1-F_access(t_crit))·h (S6.1)

Proof.

Partition at t_{crit}(h):

E_bar$_{PEP}$ = integral_0^{t_crit} E_{PEP}(t) dF + integral_{t_crit}^inf E_{PEP}(t) dF <= e_{max}·F(t_{crit}) + h·(1-F(t_{crit})),

using E_{PEP}(t) <= e_{max} for t <= t_{crit} and E_{PEP}(t) < h for t > t_{crit} (Theorem S3.1(iv)). □

Worked example: Log-normal access delay (median 72h, GSD 2.0): F(24h) ~ 0.02. With h=0.05 and e_{max}=0.95: E_bar$_{PEP}$ <= 0.02x0.95 + 0.98x0.05 = 0.068. Expected population PEP efficacy below 7%.

S7. PARAMETER TABLES

S7.1 ODE system parameters (Perelson 1996 anchored)

Symbol	Description	Value	Units	Source
d (delta)	Infected cell death rate	0.7	day^-1 (t1/2 ~1.6d)	Perelson 1996, Table 1
c	Viral clearance rate	23	day^-1 (t1/2=0.24d~6h)	Perelson 1996, Table 1
tau	Viral generation time	2.6	days (~62h)	Perelson 1996, Table 2
Eclipse phase	Min time to integration competence	0.9	days (~22h)	Perelson 1996, Table 2
beta	Infection rate constant	2.4x10^-5	mL·virion^-1·day^-1	Standard model
p	Virion production per infected cell	10^3	virions/cell/day	Standard model

alpha	Stable integration rate	10^-3	day^-1	Calibrated

PMID 8599114. Parameters c and d directly from Table 1; eclipse phase and generation time from Table 2. **CRITICAL: Reference log10 VL=4.5 for PWID is from NHBS/NHAS surveillance, NOT from Perelson 1996 (whose patients had mean log10 VL ~5.3).**

S7.2 Drug efficacy parameters

Parameter	Value	State Z(t)	Justification
e_max (pre-seeding)	0.95	Z(t)=0	Near-complete suppression of initial RT; consistent with tenofovir/FTC NHP protection data
e_mid (seeded/pre-int)	0.50	Z(t)=1	Partial suppression of established replicative focus; sensitivity: +/-40% shifts t_crit <6h
e_min (integrated)	0 (fixed)	Z(t)=2	ART does not eradicate provirus; set to 0 by A2 and Lemma S5.1

S7.3 Route-specific initial conditions and thresholds

Parameter	Mucosal	Parenteral	Note
V(0) virions/mL	1	10^3	3-log difference drives route compression; post-epithelial attenuation vs. direct IV delivery
T(0) CD4 cells/mL	10^6	10^6	Identical; peripheral blood
I(0) and R(0)	0	0	No infected cells or reservoir at exposure
I* (seeding threshold)	100 cells/mL	100 cells/mL	Sensitivity: [10,1000]; t_crit shifts <8h; compression ratio stable
R* (integration threshold)	10 cells/mL	10 cells/mL	Sensitivity: [1,100]; parenteral/mucosal ratio stable 2.5-4.0x

S7.4 Stochastic simulation parameters

Parameter	Value	Description
N Monte Carlo replications	10,000/route	Per route, per distribution, per timepoint
Time points (dense curves)	50 (0-120h)	Equally spaced; resolves all three regime boundaries
VL draws per timepoint	10,000	Independent draws from P(VL) per timepoint
Noise model (beta, alpha)	Log-normal CV=0.3	Multiplicative noise on infection and integration rates
CI type	95% credible interval	From stochastic efficacy distribution; not parameter bootstrap

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S8. STOCHASTIC VL ANALYSIS -- EXTENDED METHODS

S8.1 Source VL distributions

Expected PEP efficacy was computed as a Monte Carlo integral over source viral load:

$$E[E\_PEP(t)] = \text{integral } E\_PEP(t, VL) \cdot P(VL)\, dVL \qquad \text{(S8.1)}$$

Distribution	Mean log10	SD	Empirical basis
PWID untreated/unsuppressed	4.5	1.1	NHBS/NHAS surveillance; chronic untreated HIV-1 in PWID networks
PWID mixed treatment	3.2	1.5	Mixture: suppressed (~1.5) + unsuppressed (~4.5) in partially treated PWID networks

General HIV+ community	3.8	1.2	Mixed treatment uptake; heterogeneous MSM/general community networks
Acute-infection-enriched	5.0	0.9	High-transmission networks; acute viremia peak; recent-infection clustering

Table S8.1. All log10-normal. PWID untreated mean log10=4.5 is from NHBS/NHAS surveillance, NOT from Perelson 1996.

S8.2 Three-regime temporal classification

Regime	Hours (PWID untreat.)	Peak CI width	Clinical interpretation
1 -- Early window	0–22.0h	<15 pp	VL uncertainty low-impact; initiate PEP immediately regardless of source VL
2 -- Critical window	22.0–78.4h (PWID/acute); 22.0–88.2h (mixed/general)	39.7 pp at 49h	Maximal VL uncertainty; source VL testing clinically actionable; VL premium peaks at 21h
3 -- Late window	78.4–120h (PWID/acute); 88.2–120h (mixed/general)	<10 pp (collapsed)	Efficacy uniformly poor; VL irrelevant; P(futile) > 90%

Regime 1/2 boundary 22.04h is consistent across all four VL distributions (eclipse phase lower bound, Perelson 1996). Regime 2/3 boundary is distribution-dependent: 78.37h for PWID untreated and acute-infection-enriched distributions (higher VL accelerates CI collapse); 88.16h for mixed-treatment and general-community distributions. This distribution-specificity is biologically coherent: higher inoculum populations exhaust the critical window earlier.

S8.3 VL knowledge premium and non-monotonic plateau

VL knowledge premium = absolute difference in expected $E_{PEP}(t)$ between known vs. unknown source VL. Peak values: 3.76 pp at 21h (acute-enriched); 2.94 pp (PWID untreated); 1.87 pp (general community); 1.27 pp (mixed Tx). All distributions peak at 21h -- maximal VL-knowledge value at critical window onset.

The non-monotonic plateau between 18-36 hours in the PWID untreated and acute-enriched premium curves is a verified real kinetic signal (not numerical artefact). It arises from the interaction of the logistic inflection of the seeding-integration cascade with the 2-hour urgency reduction at the seeding midpoint in the parenteral model, creating a period during which the distribution of integration outcomes is maximally bimodal. This feature is preserved without smoothing in all reported outputs.

S9. CITY-STRATIFIED ANALYSIS -- EXTENDED METHODS

S9.1 Data source

City-level HIV surveillance data were extracted from AIDSVu downloadable datasets for 34 high-burden US metropolitan areas (2023 reporting year; downloaded October 2025).

S9.2 Mixture-model VL derivation

$$mu\_mix = f\_s \cdot mu\_suppressed + (1-f\_s) \cdot mu\_unsuppressed \qquad \text{(S9.1)}$$

Suppressed: log10 mean=1.5, SD=0.4. Unsuppressed: log10 mean=4.5, SD=1.1. IDU correction: f_s reduced by 0.12 pp per pp IDU prevalence, capped at 12 pp total.

S9.3 Structural delay derivation

$$Dt\_structural = b1 \cdot (LateDx\% - 10\%) + b2 \cdot max(90\% - Linkage\%, 0) + Dt\_SDOH \qquad \text{(S9.2)}$$

b1 = 1.2 h/%; b2 = 0.3 h/%. SDOH modifier 0-6h from Gini + poverty. Barrier categories: low (<8h), moderate (8–18h), high (18–30h), severe (≥30h).

S9.4 Counterfactual analysis

Parallel simulation assigning all 34 cities a uniform 24h access delay isolates structural barrier contribution from community VL. 33/34 cities: observed efficacy > counterfactual (structural delays <24h). Hartford CT sole exception (delay 24.4h; high-barrier category, 18–30h). Counterfactual range across cities: 76.2%–82.0%.

S10. FIGURE INVENTORY

Overview. Three main figures (Figs. 1-3), each up to four panels, generated by three Python scripts. All figures at 300 dpi. This inventory maps every panel to its source script, output file, and description.

Figure 1 -- Route-Dependent PEP Efficacy Decay

Source: `PEP_parenteral_perelson.py`, *function* `plot_vl_sweep()`. *Output:* `pep_parenteral_vl_sweep.png`

Panel	Title	Description
A	Efficacy Curves by Source Viral Load (Parenteral/PWID)	E_PEP(t) vs. hours post-exposure for 7 source VL levels (log10 = 3.0-6.0). Demonstrates VL as the primary window compressor. X-axis: 0-120h; Y-axis: efficacy 0-1. Vertical dashed lines at 24h, 48h, 72h. PRIMARY PANEL -- the money plot.
B	Biological Window Compression by Source VL	Effective prevention window (hours, defined as t_crit at eta=0.05) as a function of log10 VL. Monotone compression from ~74h (log10 VL 3.0) to ~16h (log10 VL 6.0). Horizontal reference lines at 72h (CDC) and 24h. Supports Corollary S5.1. Candidate: Supp. Fig. S1.
C	Probability PEP Already Too Late by Source VL and Time	P_int(t) as a function of log10 VL at t=24h, 48h, 72h. Shows >50% integration probability reached before 24h at high VL (log10>5). Candidate: Supp. Fig. S1.
D	Mucosal vs. Parenteral Efficacy (Route + VL Both Matter)	Paired efficacy curves for mucosal (solid) and parenteral (dashed) at matched VL values (log10=3.0, 4.5, 6.0). Demonstrates ~3-fold compression ratio. Validates NHP concordance table. PRIMARY PANEL.

For Science (4-panel allowed): include all four. If condensed, prioritize A + D; move B+C to Supp. Fig. S1.

Figure 2 -- Stochastic Efficacy Analysis Under Source VL Uncertainty

Source: `PEP_stochastic_perelson.py`*, function* `plot_stochastic_analysis()`. *Output:* `pep_stochastic_vl_uncertainty.png`

Panel	Title	Description
A	Mean + 95% CI Efficacy Curves by Community VL Distribution	Mean E_PEP(t) +/- 95% CI for all four VL distributions over 0-120h. Shaded CI bands. Regime shading overlaid: Regime 1 (white, 0–22.0h), Regime 2 (light gray, 22.0–78.4h for PWID/acute; 22.0–88.2h for mixed/general), Regime 3 (dark gray, >78.4h PWID/acute; >88.2h mixed/general) -- USE EXACT DISTRIBUTION-SPECIFIC BOUNDARIES from pep_stochastic_ci_width_regimes.csv. PRIMARY PANEL.
B	Community VL Distributions (What We Integrate Over)	Log10-normal density plots for all four VL distributions on common x-axis. Shows 1.8-log spread between extreme distributions. Provides visual grounding for Panel A. Candidate: Supp. Fig. S2.
C	Probability of Sub-Therapeutic PEP by Delay and Community	P(efficacy <50%) vs. hours post-exposure for all four distributions. Threshold line at P=0.50. Key result: P(<50%)=100% at 48h for ALL distributions. PRIMARY PANEL.
D	VL Knowledge Premium (Value of Rapid Source VL Testing)	VL knowledge premium (pp, absolute) vs. hours for all four distributions. Peaks at 21h for all distributions (3.76pp acute-enriched; 2.94pp PWID untreated). Non-monotonic plateau 18-36h PRESERVED with annotation. PRIMARY PANEL.

CRITICAL -- Panel D: The non-monotonic 18-36h plateau is a verified kinetic signal. Do NOT smooth. See S8.3. Annotation required.

CRITICAL -- Panel A: Regime boundaries are distribution-specific. R1/R2 = 22.04h for all distributions (from pep_stochastic_ci_width_regimes.csv). R2/R3 = 78.37h for PWID untreated and acute-enriched; 88.16h for mixed-treatment and general-community. Apply per-distribution shading in Panel A.

Figure 3 -- City-Stratified PEP Efficacy Across 34 US Metropolitan Areas

Source: `city_stratified_figures.py`. *Outputs:* `Fig_CityStratified_PEP.png` *(4-panel, PRIMARY) and* `Fig_CityComparison_Focus.png` *(3-panel, focus cities).*

Primary 4-panel figure:

Panel	Title	Description
A	City-Stratified PEP Efficacy at City-Specific Structural Delay	Ranked horizontal bar chart for 34 cities. Colors: green=low barrier (<8h), orange=moderate (8–18h), red=high (18–30h). Range: Milwaukee 98.3% to Hartford 78.8%. PRIMARY PANEL.
B	Structural Delay vs. PEP Efficacy (bubble = IDU prevalence %)	Scatter: x=structural delay, y=efficacy. Bubble size proportional to IDU prevalence. Color by region (South red; Northeast blue; West green; Midwest purple). Pearson r=-0.935 annotated. Key cities labeled. PRIMARY PANEL.
C	Joint VL Burden x Structural Delay Risk Surface	2D scatter/heatmap: x=community mean log10 VL, y=structural delay, color=efficacy. Shows structural delay (y) drives most variance; community VL (x) has limited effect (r=-0.084). San Juan labeled as dual-disadvantage city. Candidate: Supp. Fig. S4.
D	Actual vs. Counterfactual (24h) PEP Efficacy by City	Paired lollipop/dot: actual (solid) vs. 24h counterfactual (open) for 34 cities. 33/34 cities: actual > counterfactual. Hartford only exception (-0.59pp deficit). Counterfactual range: 76.2%-82.0%. PRIMARY PANEL.

Focus 3-panel figure (Hartford, San Juan, Jackson, Phoenix, Denver, Milwaukee):

Panel	Title	Description
A	Viral Suppression vs. Linkage to Care	Grouped bar: VS% (solid) and linkage% (faded) for 6 focus cities. Reference lines at 90% UNAIDS and US median. Shows dual compounding disadvantage.
B	Structural Delay Decomposition by Component	Stacked bar: late-dx contribution, linkage-gap contribution, SDOH modifier for 6 focus cities. Identifies dominant driver per city (Hartford=late-dx; San Juan=linkage-gap).
C	Expected PEP Efficacy (95% CI); Diamond = 24h counterfactual	Pointrange+CI for 6 focus cities with 24h counterfactual diamond overlay. Direct reading of uncertainty and counterfactual gap.

S10.1 Recommended Figure Assignment for Science Submission

Figure slot	Content	Source file	Panels
Main Fig. 1	Route-dependent window; within-host kinetics	pep_parenteral_vl_sweep.png	A + D (or all 4)
Main Fig. 2	Stochastic VL uncertainty; three clinical regimes	pep_stochastic_vl_uncertainty.png	A, C, D (Panel B to supplement)
Main Fig. 3	City-stratified structural determinants	Fig_CityStratified_PEP.png	A, B, D (Panel C to supplement)
Supp. Fig. S1	Window compression detail	pep_parenteral_vl_sweep.png	Panels B + C
Supp. Fig. S2	VL distribution densities	pep_stochastic_vl_uncertainty.png	Panel B
Supp. Fig. S3	Focus-city decomposition	Fig_CityComparison_Focus.png	All 3 panels
Supp. Fig. S4	Joint VL x structural delay risk surface	Fig_CityStratified_PEP.png	Panel C

All figures are generated by existing scripts in the project repository; no new figure generation is required for submission. Each main figure carries a self-contained argument: Fig.1 = the biological window exists and is route-specific; Fig.2 = VL uncertainty creates three clinically distinct regimes; Fig.3 = structural barriers consume the window at population scale.

Data and materials availability. All model source code, outputs, and analysis scripts are available in the project Github repository, and data are available in the project Zenodo database doi 190011771 All are open source and available for immediate use. No individual-level human data were used.

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